WORKED SOLUTION

STEP II 1990, Question 5

Pure • Complex Numbers and Circles

STEP II 1990, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q5

The key idea

On a circle $|z| = r$ conjugation is division: $\bar{z} = r^2/z$. That turns "this ratio is purely imaginary" (the test for perpendicular directions) into a neat polynomial condition, $lm + pq = 0$. After that, the rest of the question is a chase: each perpendicular chord gives the new point by a formula of the form $z \mapsto c/z$, and you just compose these maps and see what comes out.

Solution

The perpendicularity criterion

Let S have radius r, so $|l| = |m| = |p| = |q| = r$ and therefore

$$\bar{l} = \frac{r^2}{l}, \qquad \bar{m} = \frac{r^2}{m}, \qquad \bar{p} = \frac{r^2}{p}, \qquad \bar{q} = \frac{r^2}{q}.$$

The chords LM and PQ have directions $l - m$ and $p - q$, both non-zero because the points are distinct. Two non-zero complex numbers represent perpendicular directions exactly when their quotient is purely imaginary, so put

$$w = \frac{l - m}{p - q}.$$

Then w is purely imaginary if and only if $w + \bar{w} = 0$. Now

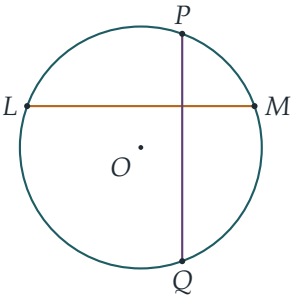
$$\bar{w} = \frac{r^2/l - r^2/m}{r^2/p - r^2/q} = \frac{(m - l)/(lm)}{(q - p)/(pq)} = \frac{l - m}{p - q} \cdot \frac{pq}{lm} = w\,\frac{pq}{lm},$$

so

$$w + \bar{w} = w\left(1 + \frac{pq}{lm}\right) = \frac{w}{lm}\,(lm + pq).$$

Since $w \neq 0$ and $lm \neq 0$, this vanishes if and only if $lm + pq = 0$. So $LM \perp PQ$ if and only if $lm + pq = 0$, as required.

(The perpendicular-bisector route is the same idea in geometric clothing: the perpendicular bisector of a chord LM passes through O, and the chord's direction is i times the direction of that bisector.)



$l = re^{i\cdot 160^\circ}$, $m = re^{i\cdot 20^\circ}$
$p = re^{i\cdot 70^\circ}$, $q = re^{-i\cdot 70^\circ}$
$lm = -r^2$, $pq = r^2$

Three points

Write a_1, a_2, a_3 for A_1, A_2, A_3, and z for A_1'. By the criterion, "$A_1'A_2' \perp A_1A_2$" means $z\,a_2' = -a_1a_2$, so

$$a_2' = -\frac{a_1a_2}{z}.$$

This point really is on S, because $|a_2'| = r \cdot r/r = r$; and it is the only point of S that works. In the same way

$$a_3' = -\frac{a_2a_3}{a_2'} = \frac{a_3}{a_1}\,z,$$
$$a_1'' = -\frac{a_3a_1}{a_3'} = -\frac{a_1^2}{z}.$$

Hence $A_1'' = A_1'$ if and only if $z = -a_1^2/z$, that is,

$$z^2 = -a_1^2 \iff z = \pm i\,a_1.$$

Both values have modulus r, so lie on S, and they are different because $a_1 \neq 0$. So there are exactly two positions of A_1': the points obtained by rotating A_1 through $\pm 90^\circ$ about O.

Ask yourself What is the map $z \mapsto -a_1^2/z$ doing to the circle?

On S we have $1/z = \bar{z}/r^2$, so $-a_1^2/z = -(a_1^2/r^2)\,\bar{z}$, which is a conjugation followed by a rotation: a reflection of the circle in a diameter. A reflection fixes exactly the two ends of its mirror, which is why there are two solutions. Indeed every single step $z \mapsto -a_ka_{k+1}/z$ is such a reflection, and that explains everything that follows.

Four points

Continuing the same chase with a_4 in place of the last step,

$$a_2' = -\frac{a_1 a_2}{z}, \qquad a_3' = \frac{a_3}{a_1} z,$$
$$a_4' = -\frac{a_3 a_4}{a_3'} = -\frac{a_1 a_4}{z}, \qquad a_1'' = -\frac{a_4 a_1}{a_4'} = z.$$

So $A_1'' = A_1'$ for every choice of A_1' on S.

n points

By induction on k, the chase gives

$$a_k' = \frac{a_k}{a_1} z \quad (k \text{ odd}), \qquad a_k' = -\frac{a_1 a_k}{z} \quad (k \text{ even}),$$

since if one form holds for a_k', then $a_{k+1}' = -a_k a_{k+1}/a_k'$ has the other form. The last step is $a_1'' = -a_n a_1 / a_n'$:

- if n is even, $a_1'' = -a_n a_1 \cdot \dfrac{z}{-a_1 a_n} = z$, so A_1'' coincides with A_1' for every A_1';
- if n is odd, $a_1'' = -a_n a_1 \cdot \dfrac{a_1}{a_n z} = -\dfrac{a_1^2}{z}$, so, exactly as for $n = 3$, A_1'' coincides with A_1' for exactly two positions of A_1', namely $z = \pm \mathrm{i} a_1$.

In words: an even number of reflections makes a rotation, which here turns out to be the identity, while an odd number makes a reflection, which has two fixed points on the circle.

Checks

- The diagram: with $l = r\mathrm{e}^{\mathrm{i}\cdot 160^\circ}$, $m = r\mathrm{e}^{\mathrm{i}\cdot 20^\circ}$ the chord LM is horizontal, and with $p = r\mathrm{e}^{\pm \mathrm{i}\cdot 70^\circ}$ the chord PQ is vertical; indeed $lm = r^2 \mathrm{e}^{\mathrm{i}\pi} = -r^2 = -pq$.
- Numerically, for random points on circles of random radius, and n from 3 to 9, chasing the perpendicular chords gave a point back on S at every step, returned exactly to the start for even n, and for odd n a scan of 200 000 starting positions round the circle found exactly two fixed points, at $\pm \mathrm{i} a_1$.

Common slips

- Testing perpendicularity by "w is real". Parallel chords give a real ratio; perpendicular ones give a purely imaginary ratio.
- Forgetting that $\bar{z} = r^2/z$ needs $|z| = r$; it is the fact that all four points are on the same circle centred at O that makes the criterion so clean.
- In the n-point case, forgetting that the answer depends on the parity of n.

Beyond STEP

The maps $z \mapsto c/z$ restricted to the circle are reflections, and composing reflections is the beginning of the study of isometry groups: the symmetries of a circle form the orthogonal group $O(2)$, in which two reflections compose to a rotation. Maps such as $z \mapsto c/z$ and $z \mapsto (az+b)/(cz+d)$ are Möbius transformations, central to complex analysis and geometry. Needham's *Visual Complex Analysis* develops both pictures beautifully.

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